Project Proposal

Grama Niladhari Digital Based Management System (GNOS)

[Group Number – 01]

AI-Integrated Mobile App Development Project

Intake 41

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Table of Contents

1. Project Description

1.1 Background 4
1.2 Summary of the Problem 4
1.3 Project Definition 5
1.4 Proposed Solution 5

2. Aim and Objectives 6

3. Scope and Limitations

3.1 Scope 6
3.2 Limitations 7

4. Data Collection Methods 7

5. Technical Approach 8

6. Requirement Specification

6.1 Functional Requirements 10
6.2 Non-Functional Requirements 10

7. Prototype User Interfaces 11

8. Monetization

8.1 Freemium Model 17

8.2 Service / Transaction Fee Model 17

8.3 Why This Feature is Important 18

8.4 Monetization Strategy 18

8.5 How It Ensures Sustainability 19

9. Repository Link 19

10. References 19

1. Project Description

1.1 Background

Grama Niladhari (GN) offices play a vital role in Sri Lanka's public administration system by providing essential citizen services such as National Identity Card (NIC) verification, person registration and issuing character certificates. These services are traditionally managed through manual and paper-based processes. Then GN officers face challenges such as excessive paperwork, time delays, human errors, difficulty in record maintenance and inefficiencies in service delivery. Also citizens, especially those in rural areas like as often experience long waiting times and repeated visits to GN offices to complete simple administrative tasks.

With the rapid growth of information technology and the government's move towards e-governance and there is a strong need to digitize these GN office operations. A digital platform can improve efficiency, transparency, accessibility and accuracy while reducing administrative burdens on officers and inconvenience to citizens. However, many GN offices still lack a unified and structured online system to manage these services effectively.

The idea for the Grama Niladhari Online Services (GNOS) project was developed after identifying these gaps in the current system. Through discussions, observations and reviewing existing administrative workflows. It became evident that a centralized mobile based system could significantly improve the management of GN services. The GNOS project is proposed as a solution to automate and streamline core GN functions while providing a public facing platform for citizens to access information and services easily.

GNOS aims to support GN officers by offering digital tools for NIC verification, person registration, character certificate issuance and Rag bot to asking questions in GN Officers and Citizens. It provides citizens with online features such as NIC validation, downloadable forms, and the ability to locate their relevant GN division and officer contact details at the same time. This project supports Sri Lanka's digital transformation goals and lays a foundation for future integration with national level government databases.

1.2 Summary of the Problem

Grama Niladhari (GN) offices in Sri Lanka mainly use manual and paper-based systems to provide services such as NIC verification, person registration and issuing character certificates. These traditional processes cause excessive paperwork, delays, human errors and difficulties in storing and retrieving records. GN officers must manage large volumes of citizen data without a structured or secure digital system, leading to inefficiencies and reduced service quality.

Citizens of specially living in rural areas they experience long waiting times and repeated visits to GN offices to complete simple administrative tasks. There is no centralized digital platform for citizens to verify NIC details, download required forms or easily identify their relevant GN

division. Then secure and centralized digital solution is required to improve efficiency, transparency, and accessibility of GN services.

1.3 Project Definition

The primary problem addressed by this project is the inefficiency of manual administrative processes in Grama Niladhari offices. GN officers are responsible for handling large volumes of citizen data, including identity verification, population records and certification. Currently in these records are often maintained in physical files and basic digital formats without proper structure and security.

Manual handling of records leads to several issues such as data loss, duplication, difficulty in retrieval, and higher chances of human error. Processing requests such as character certificates or person registration takes considerable time, affecting service quality and public satisfaction. Citizens must physically visit GN offices multiple times, which is inconvenient and costly, especially for those living in remote areas.

Other reason is there is no centralized digital platform for citizens to verify NIC details, download required forms and easily identify their respective GN division. The lack of digital records also limits transparency and accountability in administrative services.

There is a critical need for a secure, centralized and user-friendly digital system that supports GN officers in managing their duties efficiently while improving service accessibility for citizens.

1.4 Proposed Solution

The proposed solution is a mobile based Digital Service Management System named GNOS. This system digitizes the key administrative services offered by Grama Niladhari offices. GNOS provides a secure login system for GN officers and allowing them to perform NIC verification, person registration and character certificate issuance through structured digital forms.

The system uses modern mobile technologies for the frontend and cloud-based backend services to ensure reliability and scalability. All data is securely stored and managed using Firebase databases that ensuring real time updates and data integrity. A citizen portal is included in enabling users to validate NIC numbers, download official forms and locate GN division details without visiting offices physically.

This project includes a dual-interface Retrieval Augmented Generation (RAG) system to support and digitize Grama Niladhari (GN) services in Sri Lanka. The citizen facing module provides instant and accurate information on administrative procedures, document requirements and service eligibility, reducing repeated visits to GN offices and lowering officer workload. The GN officer module acts as a decision support tool, allowing officers to query official government circulars, gazettes and directives to ensure procedural and legal accuracy.

GNOS reduces paperwork, improves processing speed, enhances data accuracy and increases transparency. It also acts as a foundation for future improvements such as government database integration in SMS or Email notifications and expanded GN services.

2. Aim and Objectives

Aim

To develop a mobile based digital system that is AI automates and improves Grama Niladhari administrative services for enhanced efficiency, accuracy and accessibility.

Objectives

- To design and implement a secure mobile based with artificial intelligence platform for GN officers.
- To develop a RAG bot to GN officers and Citizens for asking some essential questions.
- To digitize NIC verification, person registration and character certificate issuance.
- To create a centralized database for GN administrative records.
- To provide citizens with online access to NIC validation, downloadable forms and GN office information.
- To reduce manual paperwork and processing time.
- To improve transparency and reliability in GN service delivery.

3. Scope & Limitations

3.1 Scope

- GN officer registration and authentication.
- NIC verification module.
- Person registration module.
- Character certificate issuance module.
- RAG bot to answering to questions.
- Secure storage of records using cloud databases.
- Citizen portal for NIC validation and form downloads.
- GN division and officer contact search functionality.

3.2 Limitations

- No real time SMS or email notifications.
- Requires stable internet access.
- Designed as a mobile application only not in the web application.

4. Data Collection Methods

Primary Data Collection Methods

- Grama Niladhari officers
- Citizens

Secondary Data Collection Methods

- Government circulars
- Gazettes
- Administrative guidelines
- Official GN forms
- Policy documents

Data Collection for AI (RAG Bot)

- Curated GN documents converted to text
- Indexed into vector database
- Only approved and non-sensitive documents used
- No personal data used for model training

5. Technical Approach

❖ System Architecture Overview

The Grama Niladhari Online Services (GNOS) system follows a modular and mobile-based architecture designed to ensure scalability, security and real time responsiveness. The architecture separates concerns into frontend, backend services, databases, authentication and AI components.

The system is built using Flutter (Dart) for the frontend, while Firebase services are used for authentication, real-time data handling and secure storage. An AI-powered Retrieval-Augmented Generation (RAG) chatbot is integrated to assist both Grama Niladhari officers and citizens.

❖ Frontend Layer (Mobile Application)

- Developed using Flutter (Dart) for cross platform compatibility.
- Modular UI components handle:
    a. User authentication
    b. NIC verification
    c. Person registration
    d. Certificate services
    e. AI chatbot interaction

❖ Authentication & Authorization

- Firebase Authentication is used for secure login and user management.
- Supports:
    a. Email & password authentication
    b. Secure session handling
- Implements Role Based Access Control
    a. GN Officers: Full access to administrative modules.
    b. Citizens: Limited access in NIC validation and downloads

❖ Data Storage & Real-Time Processing

- Firebase Realtime Database
    - Stores dynamic and frequently updated data.
    - Enables real time synchronization across devices.
- Firestore Database
    - Stores structured and long-term data such as:
        a. Officer profiles
        b. Citizen records
        c. Certificate metadata.
- Firebase Security Rules
    a. Enforce access control
    b. Prevent unauthorized data access
    c. Ensure data integrity

❖ AI Integration: RAG Based Chatbot

The GNOS system integrates a RAG chatbot to provide accurate and context aware responses to both Grama Niladhari officers and citizens.

- RAG Workflow
    a. User submits a question via the Flutter chatbot interface.
    b. The query is converted into vector embeddings.
    c. Relevant documents are retrieved from the vector database.
    d. Retrieved content is passed to a Large Language Model.
    e. The LLM generates an accurate response grounded in retrieved GNOS data.
    f. The answer is returned to the mobile app in real time.

❖ AI Use Cases in GNOS

- Citizen assistance.
- GN officer support.
- Multilingual Q&A
- Reduced workload and faster service delivery.

❖ Security & Data Privacy

- All communications use HTTPS and Firebase security layers.
- Sensitive data (NIC numbers, personal details) is:
    a. Access-restricted by role
    b. Never used directly in AI model training
- AI responses are generated only from approved and non-sensitive documents.

❖ Real-Time Updates & System Reliability

- Real-time updates are achieved through Firebase Realtime Database listeners.
- Instant UI updates for:
    a. Registration status
    b. Certificate generation
    c. Data changes

6. Requirement Specification

6.1 Functional Requirements

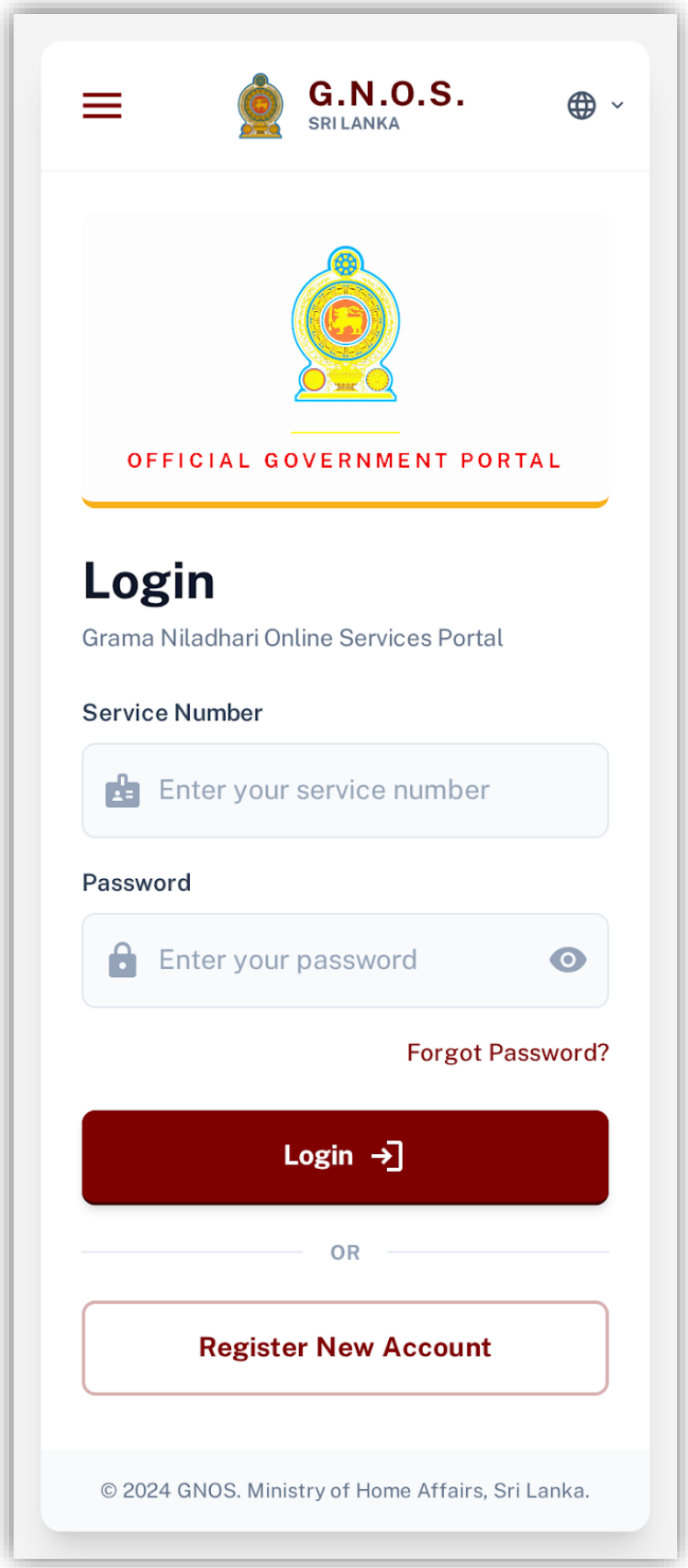
- User registration and login
- RAG Bot to Answering
- NIC verification
- Person registration
- Certificate generation and download
- Citizen portal access
- Data storage and retrieval

6.2 Non-Functional Requirements

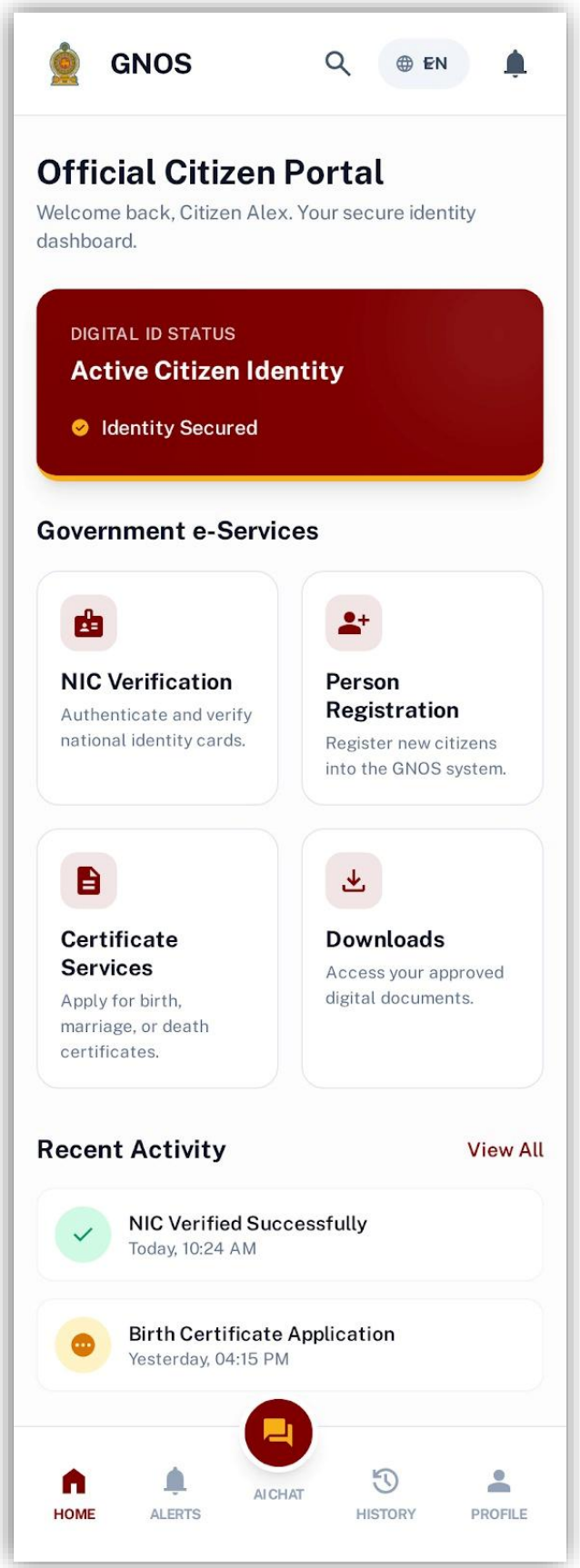
- Security and authentication
- Performance and responsiveness
- Usability and multilingual support
- Reliability and scalability
- Data integrity and backup

7.Prototype User Interfaces

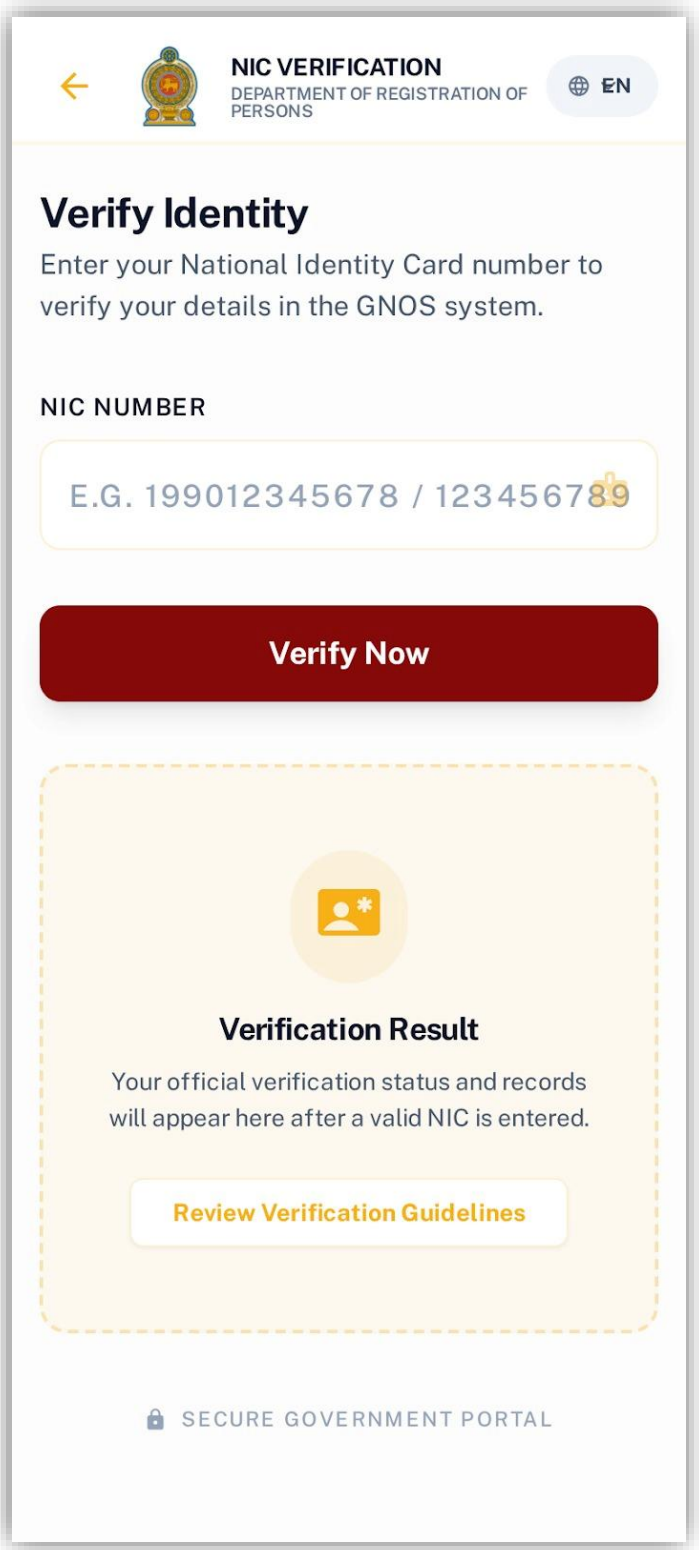
01. Loging Page



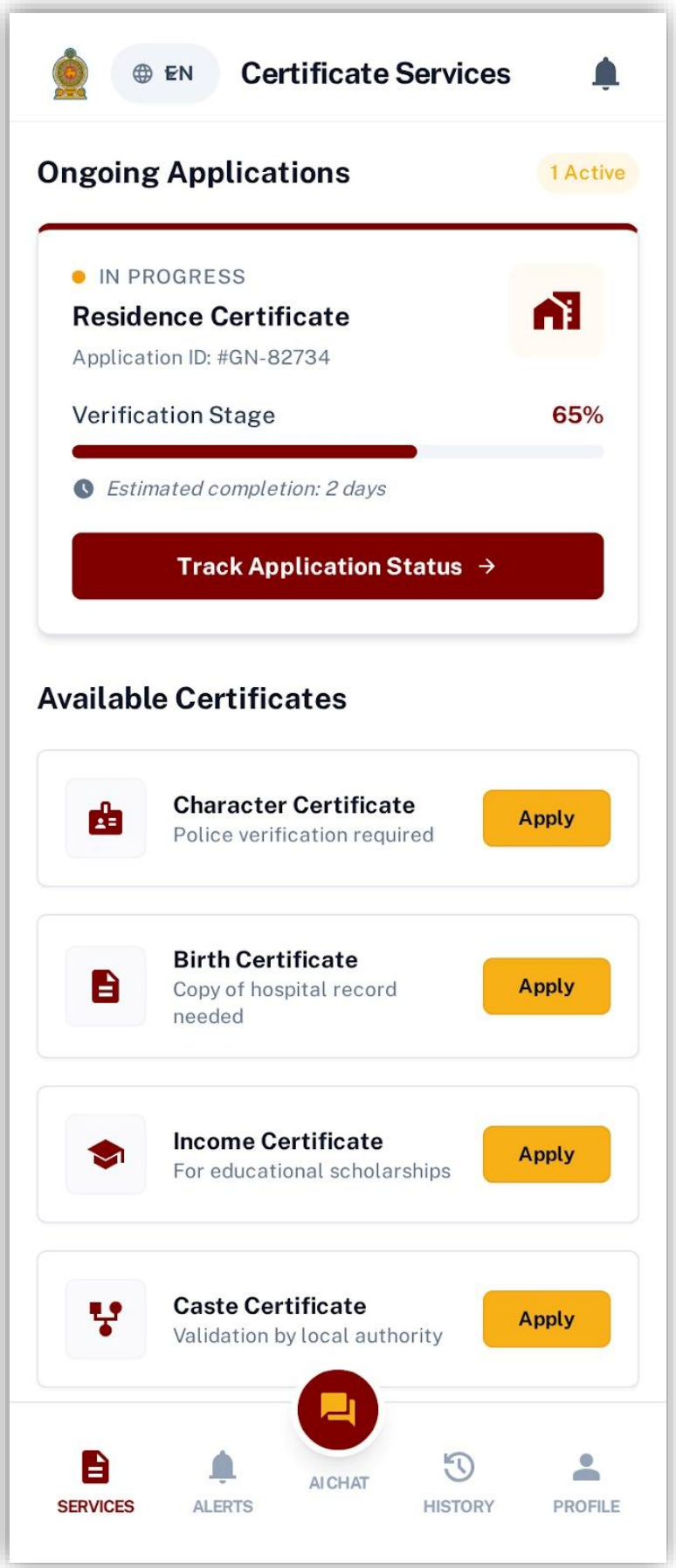
02. Home Page


GNOS
EN
Official Citizen Portal
Welcome back, Citizen Alex. Your secure identity dashboard.
DIGITAL ID STATUS
Active Citizen Identity
Identity Secured
Government e-Services
NIC Verification
Authenticate and verify national identity cards.
Person Registration
Register new citizens into the GNOS system.
Certificate Services
Apply for birth, marriage, or death certificates.
Downloads
Access your approved digital documents.
Recent Activity
View All
NIC Verified Successfully
Today, 10:24 AM
Birth Certificate Application
Yesterday, 04:15 PM
HOME
ALERTS
AI CHAT
HISTORY
PROFILE

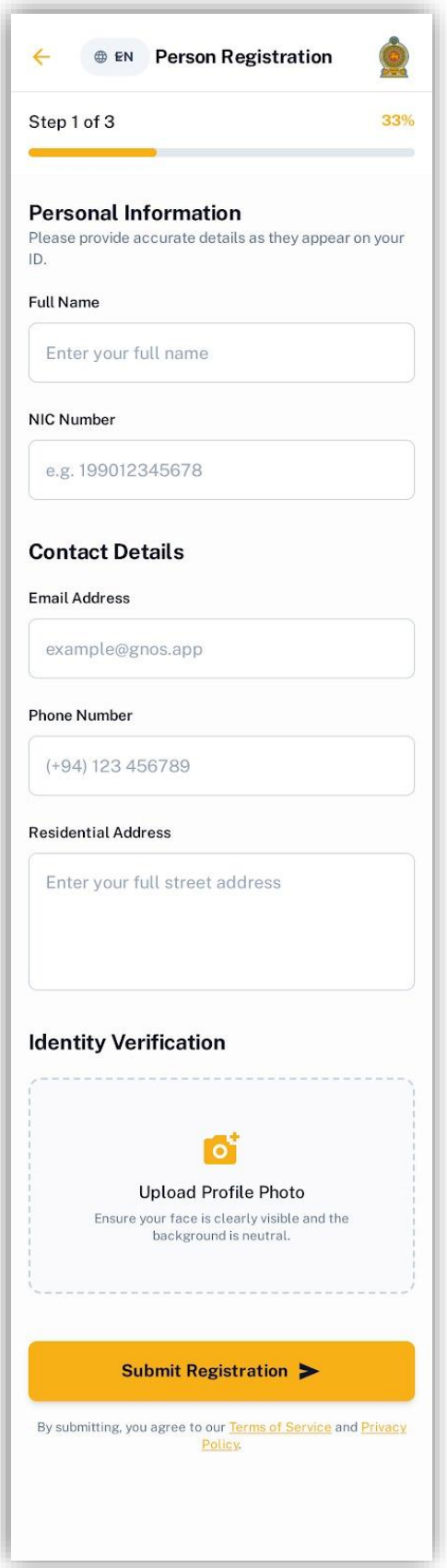
03. NIC Verification



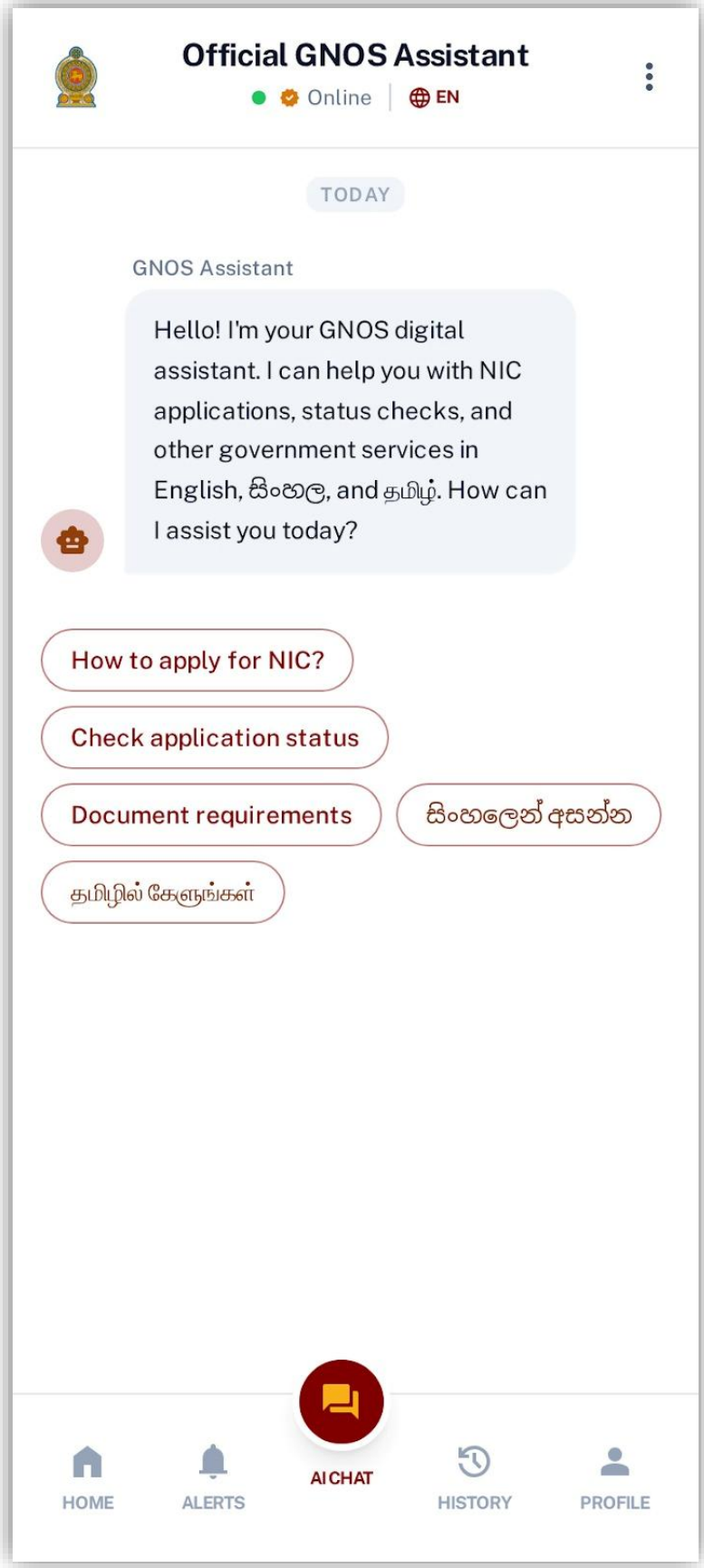
04. Official Certification Service



05. Official Person Registration Service



06. Official GNOS Assistant



8.Monetization

The GNOS mobile application will adopt a Freemium Model combined with Government Partnership and Service Transaction Fees.

8.1. Freemium Model

This is available for free to all citizens.

- Basic services such as:
    - NIC validation
    - GN division search
    - Downloadable forms
    - General AI chatbot support

8.2. Service / Transaction Fee Model

A small service charge can be applied for:

Feature Idea: Local Village Marketplace with Small Commission Model

Feature Description - The application will include a Village Marketplace Module that allows small and medium-scale businesses within the village to sell their products directly to residents through the mobile app.

This feature enables:

- Local entrepreneurs to list their products
- Villagers to browse and purchase products within their own GN division
- Digital payment processing through the app

When a customer purchases a product, a small commission fee will be automatically deducted by the application from the total transaction amount.

How It Works

1. Small and medium-scale business owners register in the app as sellers.
2. They upload product details such as:
    - Product name
    - Price
    - Description
    - Photos

3. Villagers can browse products available in their local area.
4. When a purchase is made:
    - The buyer pays through the app.
    - A small percentage (for example 3%–5%) is automatically deducted as a service fee.
    - The remaining amount is transferred to the seller.

8.3. Why This Feature is Important

1. Supports Local Economy

- Encourages villagers to buy from local businesses.
- Helps small-scale entrepreneurs increase their income.
- Keeps money circulating within the village economy.

2. Promotes Digital Payments

- Reduces cash handling.
- Provides secure transaction records.
- Increases financial transparency.

3. Convenience for Villagers

- Easy access to locally made products.
- Saves transportation time and cost.
- Builds stronger community connections.

8.4. Monetization Strategy

The app will generate revenue through a Transaction Commission Model,

- A small percentage fee will be deducted from each successful transaction.
- This ensures that,

    - Sellers do not pay large upfront fees.

    - The app earns revenue only when transactions happen.

    - The system remains fair and sustainable.

8.5. How It Ensures Sustainability

The commission-based model ensures,

- Continuous income for app maintenance
- Coverage of hosting and database costs
- Funding for future feature development
- Long-term financial stability of the application

Example Scenario

If a villager buys homemade food products worth Rs. 2,000,

- 5% commission = Rs. 100
- Seller receives Rs. 1,900
- Rs. 100 goes to app maintenance and system sustainability

9. Repository Link

thisass/GNOS.

10.References

1. United Nations, *E-Government Survey 2022: The Future of Digital Government*, New York, NY, USA: United Nations, 2022

2. Government of Sri Lanka, *Digital Government Policy Framework*, Ministry of Technology, Colombo, Sri Lanka, 2020.

3. Al-Bar and M. Hoque, "Web-based e-government systems: A review," *International Journal of Information Management*, vol. 37, no. 2, pp. 111–119, Apr. 2017.

4. Al-Bar and M. Hoque, "Web-based e-government systems: A review," *International Journal of Information Management*, vol. 37, no. 2, pp. 111–119, Apr. 2017.

5. Sommerville, *Software Engineering*, 10th ed., Boston, MA, USA: Pearson, 2016.

6. Google, *Firebase Documentation*, Google LLC, 2024. [Online]. Available: https://firebase.google.com/docs

7. M. Kleppmann, *Designing Data-Intensive Applications*, Sebastopol, CA, USA: O'Reilly Media, 2017.

8. P. Lewis, E. Perez, A. Piktus, et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, Vancouver, Canada, 2020, pp. 9459–9474.